

1 Feature Extraction (100pts)

In this section, you should perform bag-of-words-feature + SVM classification on a given dataset by completing [bonus_code.py](#). Before that, you need to download the CIFAR dataset from <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz> and put the extracted files in a folder named `cifar-10-batches-py`. First, build a dictionary based on the training images by using the function `build_center` (Line 232). You need to complete the code of `calcSiftFeature` and `learnVocabulary` in `build_center`. Specifically, for `calcSiftFeature`, you need to use the `sift.detectAndCompute` to compute the SIFT feature. Please refer to https://docs.opencv.org/4.x/da/df5/tutorial_py_sift_intro.html and fill in the missing line 25. As for `learnVocabulary`, it uses the K-Means algorithm `cv2.kmeans` to create a dictionary for training images. Please fill in the missing line 59. Second, leverage the function `cal_vec` to extract SIFT features for training images. Third, train a SVM classifier. You need to set a SVM classifier using `svm.SVC`. You can refer to <https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html> and fill in the missing line 159. Finally, the skeleton code will conduct the testing procedure.

Note that:

1) Line 6 in the skeleton code (`!pip install opencv-contrib-python==4.3.0.36`) is to re-install OpenCV in Colab, since the default version of OpenCV in Colab doesn't contain the SIFT algorithm. After re-installing OpenCV, you need to restart Runtime of Colab. Then you can use the new version of OpenCV.

2) Following the skeleton code, the obtained accuracy should be above 75%. You can fine-tune any parts to achieve a higher performance (feature extractor or SVM classifier).

Note: the source code should be attached with your submission.